

Mathematical and Computational Methods

Exercise Sheet 9: Integration I: the definite integral and substitution

This sheet accompanies *Unit 9 (Integration I: the definite integral and substitution)*. It exercises Riemann sums and signed area, both parts of the Fundamental Theorem of Calculus, and integration by substitution; no technique from Unit 10 (parts, partial fractions, trigonometric substitution, improper integrals) is needed.

How to use this sheet. Problems are graded by difficulty—★ drill (routine fluency), ★★ core (standard, tutorial level) and ★★★ challenge (harder or extension)—and organised in three sections: **warm-up drills** for fluency, **tutorial problems** (the core set for the 2-hour class, with the live items flagged ►) and **further practice** for independent home study.

Solutions. This is the *student* sheet (problems only); a full worked solution sits after each problem but is hidden. To produce the solutions version, uncomment `\solutionson` in the preamble (or build with `pdflatex "\def\showssolutions{}\input{MCM_Exercises_Unit9}"`) and recompile.

Toolkit. Riemann sums and signed area; the power rule and the table of standard integrals; both parts of the Fundamental Theorem of Calculus (FTC); and integration by substitution, including changing the limits and the $f'/f \rightarrow \ln$ pattern.

1 Warm-up drills

Short, single-idea problems — at least one for every skill in the unit.

Riemann sums and signed area

Exercise 1.1 (★ drill — a right-endpoint Riemann sum). Estimate the area under $f(x) = x^2$ on $[0, 2]$ using $n = 4$ rectangles of equal width and the height at the *right* end of each strip. Is your estimate an over- or under-estimate, and why?

Exercise 1.2 (★ drill — the sign of an integral). Without computing any antiderivative, decide the sign (positive, negative or zero) of each definite integral, and say why.

$$(a) \int_{-2}^2 x^3 \, dx, \quad (b) \int_0^1 e^{-x} \, dx, \quad (c) \int_{\pi}^{2\pi} \sin x \, dx.$$

Exercise 1.3 (★ drill — sketching the rectangles). A picture first.

- Sketch $y = x^2$ on $[0, 2]$ and, on the same axes, draw the four right-endpoint Riemann rectangles of Exercise 1 (width 0.5). Mark whether the rectangles lie above or below the curve.
- On separate axes sketch $g(x) = x - 1$ on $[0, 2]$. Shade the region between the graph and the x -axis, marking the part that counts as *negative* signed area. Then compute the net signed area $\int_0^2 (x - 1) \, dx$ and the total (unsigned) area.

Properties of the definite integral

Exercise 1.4 (*★ drill — properties of the definite integral*). Suppose $\int_0^2 f = 5$ and $\int_2^5 f = -2$. Using only the properties of the definite integral (do *not* find f), evaluate

$$(a) \int_0^5 f, \quad (b) \int_5^2 f, \quad (c) \int_2^2 f, \quad (d) \int_0^2 3f, \quad (e) \int_5^0 f.$$

Indefinite integration and the power rule

Exercise 1.5 (*★ drill — a polynomial antiderivative*). Find $\int (x^3 - 4x + 2) \, dx$.

Exercise 1.6 (*★ drill — powers of x first*). Write each integrand as a power of x first: $\int \left(\sqrt{x} + \frac{1}{x^2} \right) \, dx$.

Exercise 1.7 (*★ drill — standard integrals*). Use the table of standard integrals: $\int \left(2e^x - \frac{3}{x} + 5 \cos x \right) \, dx$.

Standard integrals (including arctan/arcsin)

Exercise 1.8 (*★ drill — standard forms with a linear inside*). Evaluate the indefinite integrals

$$(a) \int e^{3x} \, dx, \quad (b) \int \sec^2(2x) \, dx, \quad (c) \int \frac{dx}{9+x^2}, \quad (d) \int \frac{dx}{\sqrt{4-x^2}}, \quad (e) \int \cos(5x) \, dx.$$

FTC Part 2: evaluating definite integrals

Exercise 1.9 (*★ drill — definite integrals by FTC 2*). Evaluate

$$(a) \int_1^3 x^2 \, dx, \quad (b) \int_0^{\pi/2} \cos x \, dx, \quad (c) \int_1^e \frac{dx}{x}, \quad (d) \int_0^1 e^{2x} \, dx.$$

FTC Part 1: differentiating an integral

Exercise 1.10 (*★ drill — FTC Part 1*). Differentiate with respect to x (do *not* try to evaluate the integral):

$$(a) \frac{d}{dx} \int_0^x e^{-t^2} \, dt, \quad (b) \frac{d}{dx} \int_2^x \sin(t^3) \, dt, \quad (c) \frac{d}{dx} \int_1^x \frac{dt}{1+t^4}.$$

Substitution (reverse chain rule and f'/f)

Exercise 1.11 (*★ drill — substitution*). Each integrand contains an “inside” function whose derivative is present. Integrate by substitution:

$$(a) \int 2x(x^2+1)^5 \, dx, \quad (b) \int \cos x e^{\sin x} \, dx, \quad (c) \int 3x^2(x^3+7)^4 \, dx.$$

Exercise 1.12 (*★ drill — the f'/f pattern*). Spot the f'/f pattern and write down the logarithm:

$$(a) \int \frac{2x}{x^2+5} \, dx, \quad (b) \int \frac{\cos x}{\sin x} \, dx, \quad (c) \int \frac{e^x}{e^x+1} \, dx.$$

Exercise 1.13 (*★ drill — changing the limits*). Evaluate $\int_0^1 x e^{x^2} \, dx$ by changing the limits to the new variable.

No elementary antiderivative

Exercise 1.14 (*★ drill — elementary antiderivatives?*). For each function, state whether it has an elementary antiderivative.

$$(a) e^{-x^2}, \quad (b) x e^{-x^2}, \quad (c) \frac{\sin x}{x}, \quad (d) \sin(x^2), \quad (e) x \cos(x^2).$$

2 Tutorial problems

The core set for the two-hour tutorial. Standard difficulty, anchored on the worked examples of Unit 9, with one problem flavoured for each cohort. Problems flagged ► are the live core worked together in the two-hour session; the remaining problems are for completion at home.

Exercise 2.1 (*★★ core — Riemann sums against the exact area*). ► Let $f(x) = 1 + x^2$ on $[0, 3]$, cut into $n = 3$ strips of width 1.

- Compute the right-endpoint and left-endpoint Riemann sums.
- Using FTC Part 2, find the exact area and check it lies between your two estimates.

Exercise 2.2 (*★★ core — simplify, then integrate*). ► Integrate, simplifying first where it helps:

$$(a) \int \frac{x^2 - 3x + 2}{x} dx, \quad (b) \int (e^{2x} + \frac{3}{x} - \cos 5x) dx.$$

Exercise 2.3 (*★★ core — FTC evaluation*). (*FTC evaluation — worked-example anchor*). ► Evaluate $\int_2^4 x^2 dx$ and $\int_0^1 (3x^2 - 4x + 5) dx$, and state in one sentence what the first number represents geometrically.

Exercise 2.4 (*★★ core — a rate of flow (physics)*). [Physics / applied] Water flows into a tank at the rate $r(t) = 6t$ litres per minute, $0 \leq t \leq 5$. The volume added between times a and b is $\int_a^b r(t) dt$.

- How much water enters during the first 5 minutes?
- During which single minute $[t, t + 1]$ does the most water enter, and how much?

Exercise 2.5 (*★★ core — displacement and distance (physics)*). [Physics / applied] A particle moves on a line with velocity $v(t) = 3t^2 - 12t + 9$ (m s⁻¹) for $0 \leq t \leq 3$.

- Find the net displacement $\int_0^3 v dt$.
- Find the total distance travelled, $\int_0^3 |v| dt$.

Exercise 2.6 (*★★ core — substitution with a limit change*). (*Substitution with a limit change — worked-example anchor*). ► Evaluate $\int_0^2 x\sqrt{x^2 + 1} dx$ by converting the limits to the new variable.

Exercise 2.7 (*★★ core — a stray x to rewrite*). Evaluate $\int_0^3 x\sqrt{x+1} dx$. (Here the substitution leaves a stray x behind — rewrite it in terms of u .)

Exercise 2.8 (*★★ core — two routes to a logarithm*). ► Evaluate two integrals, contrasting the methods:

$$(a) \text{ by the } f'/f \text{ pattern, } \int \tan x dx, \quad (b) \text{ by substitution, } \int_1^2 \frac{\ln x}{x} dx.$$

Exercise 2.9 (★★ core — completing the square). ▶ By completing the square, evaluate $\int \frac{dx}{x^2 + 2x + 2}$.

Exercise 2.10 (★★ core — a probability density (data science)). [Data science / probability] A continuous random variable X has probability density $f(x) = kx(1-x)$ on $[0, 1]$ and $f(x) = 0$ elsewhere. A density must satisfy $\int_0^1 f = 1$, and $\mathbb{P}(X \leq c) = \int_0^c f$ (probability is area).

- Find k .
- Find $\mathbb{P}(X \leq \frac{1}{2})$ and comment on the answer.

Exercise 2.11 (★★ core — FTC 1 with the chain rule). [Mathematics / pure] ▶ Let $g(x) = \int_0^{x^2} \cos t \, dt$. Find $g'(x)$ in two different ways and check they agree.

Exercise 2.12 (★★ core — odd and even on a symmetric interval (maths)). [Mathematics / pure] ▶ Let f be continuous on $[-a, a]$. Using the substitution $u = -x$ on $\int_{-a}^0 f$, prove that $\int_{-a}^a f(x) \, dx = 0$ whenever f is odd, and that it equals $2 \int_0^a f(x) \, dx$ whenever f is even.

Exercise 2.13 (★★ core — spot the error). (Spot the error / conceptual.)

- A student writes $\int_{-1}^1 \frac{dx}{x^2} = \left[-\frac{1}{x}\right]_{-1}^1 = (-1) - (1) = -2$. The integrand is positive everywhere it is defined, so a negative “area” is impossible. What went wrong?
- True or false: “ $\int_a^b f(x) \, dx$ is a function of x .” Justify.

3 Further practice (home)

A larger bank for independent study, running from routine to harder. Challenge and extension problems are marked ★ ★ ★.

Indefinite integrals

Exercise 3.1 (★ drill — indefinite drill). Find

- $\int (4x^3 - 6x^2 + 1) \, dx$, (b) $\int \left(\frac{2}{x^3} + 5\sqrt{x}\right) \, dx$, (c) $\int \left(e^{-x} + \frac{1}{1+x^2}\right) \, dx$, (d) $\int (3 \sin x - 4 \sec^2 x) \, dx$.

Definite integrals (FTC Part 2)

Exercise 3.2 (★ drill — definite drill). Evaluate

- $\int_1^2 \frac{dx}{x^2}$, (b) $\int_0^{\ln 2} e^x \, dx$, (c) $\int_0^{\pi/4} \sec^2 x \, dx$, (d) $\int_{-1}^1 (x^3 + x) \, dx$, (e) $\int_0^4 \sqrt{x} \, dx$.

Substitution — indefinite

Exercise 3.3 (★ drill — substitution drill). Integrate

- $\int x^2(x^3 + 1)^7 \, dx$, (b) $\int \sin^4 x \cos x \, dx$, (c) $\int \frac{(\ln x)^2}{x} \, dx$, (d) $\int x e^{-x^2} \, dx$.

Substitution — definite**Exercise 3.4** (★★ core — *definite substitution*). Evaluate

$$(a) \int_1^2 \frac{\ln x}{x} dx, (b) \int_0^{\pi/2} \sin x \cos x dx, (c) \int_0^1 \frac{x}{(1+x^2)^2} dx, (d) \int_0^1 \frac{2x}{x^2+1} dx.$$

The $f'/f \rightarrow \ln$ pattern**Exercise 3.5** (★★ core — *the f'/f pattern*). Integrate

$$(a) \int \frac{3x^2+2}{x^3+2x+1} dx, \quad (b) \int \frac{\sec^2 x}{\tan x} dx.$$

arctan and arcsin targets**Exercise 3.6** (★★ core — *inverse-trig standard forms*). Evaluate the definite integrals

$$(a) \int_0^{\sqrt{3}} \frac{dx}{1+x^2}, \quad (b) \int_0^1 \frac{dx}{\sqrt{4-x^2}}, \quad (c) \int_0^2 \frac{dx}{4+x^2}.$$

Substitution leaving an x behind**Exercise 3.7** (★★ core — *a substitution leaving a stray x*). Find $\int \frac{x}{\sqrt{x+4}} dx$.**Signed area versus total area****Exercise 3.8** (★★ core — *signed area and total area*). For each function find both the net signed area and the total (painted) area over the stated interval.

- (a) $\sin x$ on $[0, 2\pi]$.
 (b) $x^2 - 1$ on $[0, 2]$.

Integral equations and applications**Exercise 3.9** (★★ core — *solve for the limit*). Find the positive value of a for which $\int_0^a (2x+1) dx = 12$.**Exercise 3.10** (★★ core — *work done on a spring (physics)*). [Physics / applied] A spring obeys Hooke's law with force $F(x) = kx$ ($k = 200 \text{ N m}^{-1}$), where x is the extension. The work done stretching it from $x = 0$ to $x = d$ is $W = \int_0^d F(x) dx$. Find the work to stretch the spring by 0.1 m.**Exercise 3.11** (★★ core — *a triangular density (data science)*). [Data science / probability] X has the triangular density $f(x) = \frac{1}{2}x$ on $[0, 2]$ (and 0 elsewhere).

- (a) Verify it is a valid density.
 (b) Find $P(X \leq 1)$ as an area.

Challenges and extensions

Exercise 3.12 (***challenge — FTC 1 with the chain rule). Use FTC Part 1 together with the chain rule to differentiate

$$(a) \frac{d}{dx} \int_0^{x^2} e^t dt, \quad (b) \frac{d}{dx} \int_x^{x^2} \ln(1+t^2) dt.$$

Exercise 3.13 (***challenge — completing the square, log plus arctan). A linear numerator over an irreducible quadratic splits into a logarithm plus an arctangent. Evaluate $\int \frac{x+1}{x^2+4x+5} dx$. (Hint: complete the square below, and write the numerator as a multiple of the denominator's derivative plus a constant.)

Exercise 3.14 (***challenge — an odd power under a root). Evaluate $\int \frac{x^3}{\sqrt{x^2+1}} dx$. (Hint: $u = x^2 + 1$, and write $x^3 = x^2 \cdot x$.)

Exercise 3.15 (***challenge — spot the error). (Spot the error.) A student evaluates $\int_0^{\pi/2} \sin^2 x \cos x dx$ thus:

$$\text{“set } u = \sin x; \int_0^{\pi/2} u^2 du = \left[\frac{1}{3} u^3 \right]_0^{\pi/2} = \frac{1}{3} \left(\frac{\pi}{2} \right)^3 \text{.”}$$

Identify both mistakes and give the correct value.

Exercise 3.16 (***challenge — no elementary antiderivative (data science)). [Data science / numerical] The integral $\int_0^1 e^{-x^2} dx$ has no elementary antiderivative (it gives a value of the error function).

- Explain why no FTC 2 “plug-in” will evaluate it exactly.
- Bracket its value using left- and right-endpoint Riemann sums with $n = 2$ ($\Delta x = \frac{1}{2}$). Is the true value inside your bracket?